ICT 1201

Fundamentals of Computer Systems

Introduction to Application Software

What is a Software ?

- The set of electronic program instructions or data a computer processor reads in order to perform a task or operation.
- Divided into two main categories.
  - System Software
  - Application Software

System Software ?

- System software is a type of computer program that is designed to run a computer's hardware and application programs.
- If we think of the computer system as a layered model, the system software is the interface between the hardware and user applications.
- Without systems software installed in our computers we would have to type the instructions for every thing we wanted to the computer to do.

Examples of System Software

- Operating Systems
  - It is a software program that enables the computer hardware to communicate and operate with the software.
  - E.g. Microsoft Windows, Apple MacOS, Ubuntu, Android, ios

Examples of System Software

- The BIOS (basic input/output system)
  - Gets the computer system started after you turn it on.
  - Manages the data flow between the operating system and attached devices such as hard disk, keyboard, mouse and printer.

Examples of System Software

- The boot program
  - Loads the operating system into the computer's main memory (RAM)

- Assembler
  - Take basic computer instructions and converts them into a pattern of bits that the computer's processor can use to perform its basic operations.

Examples of System Software

- Device driver
  - Controls a particular type of device that is attached to your computer, such as a keyboard or a mouse.

- Utility Software
  - Indented to analyze, configure, monitor, or help maintain a computer.
  - E.g. Antivirus, Backup software, compression utility, Debuggers, Disk checkers.

Application Software

- Often called productivity programs or end-user programs because they enable the user to complete tasks, such as creating documents, spreadsheets, databases, sending email, designing graphics and even play games.
- Application software is specific to the task it is designed.

Application Software

- Can be divided into two categories.
  - Basic applications (General purpose applications)
  - Specialized applications (Special purpose applications)

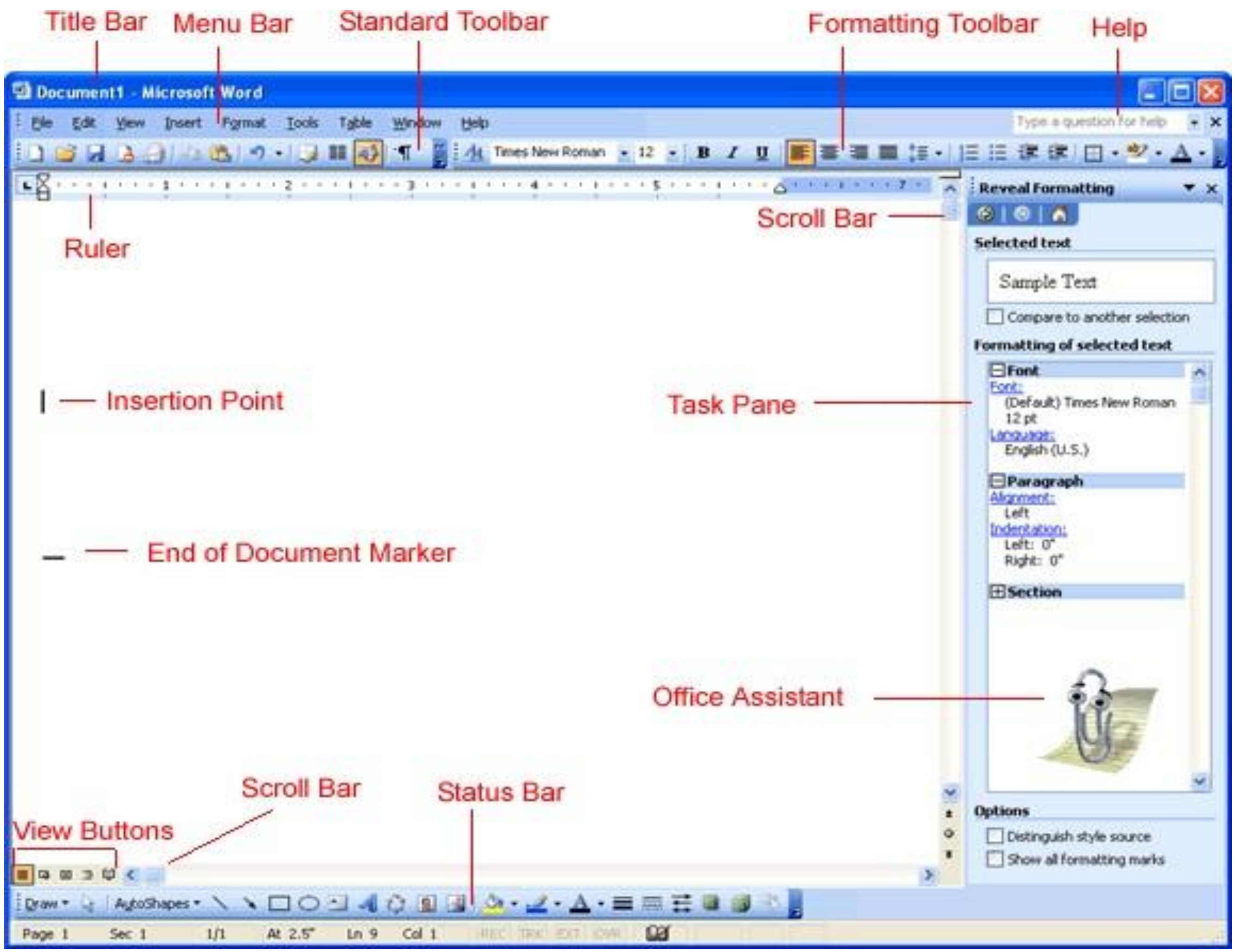
Basic Applications (General Purpose Applications)

- Widely used in nearly every discipline and occupation.
- E.g. Browsers, Word processors, Spreadsheets, Database management systems, Presentation software, etc.

Specialized Applications (Special Purpose Applications)

- Focused on specific disciplines and occupations.
- E.g. Graphics programs, Audio/video editors, Virtual reality, etc.

Common Features



Common Features

- User Interface
  - Graphical User Interface (GUI) displays information in windows and it has icons, pointers, windows, menus, menu bar, dialog box, toolbars and other buttons.

Common Features

- Window
  - A rectangular area that can contain a document, message of program.
  - More than one window can be opened at one time and can be resized, moved and closed.

- Menus
  - Have commands and these are displayed in a menu bar at top of the screen.
  - Selecting a menu item displays a pull down menu which displays more commands or options.

Common Features

- Help
  - Provide features like a keyword index and a search feature to locate information needed.

- Toolbars
  - Below the menu bar containing buttons and menus that provide quick access or shortcuts to commonly used commands.
  - Most common toolbars are standard and formatting toolbar.

Word Processing Software

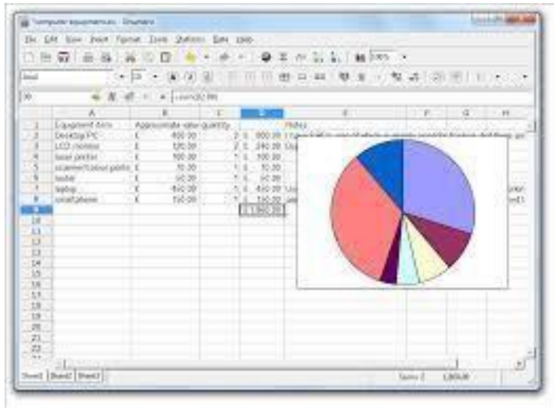
- Used to manipulate a text document, such as resume or a report.
- You typically enter text by typing and the software provides tools for copying, deleting and various types of formatting.

Some of the functions of word processing software

- Creating, editing, saving and printing documents.
- Copying, pasting, moving and deleting text within a document.
- Formatting text, such as font type, bolding, underlining or italicizing.
- Creating and editing tables.
- Inserting elements from other software, such as illustrations or photographs.
- Correcting spelling and grammar.

Electronic Spreadsheets

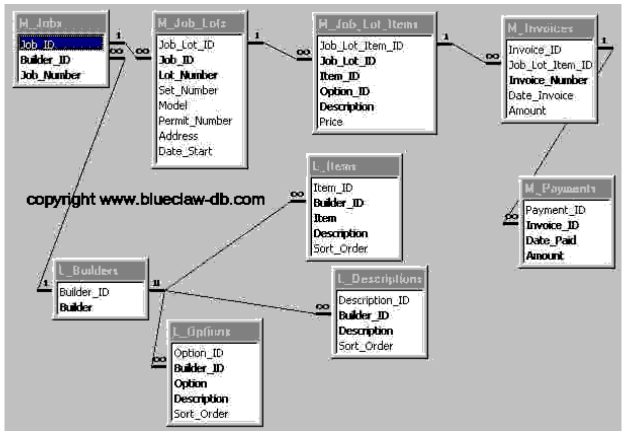
- Organizations use large amount of data. Two types of software applications are particularly suited to work with data,
  - Spreadsheet software
  - Database software



- Spreadsheet software is used to organize and manipulate numerical data.
  - Numbers are organized on a grid of lettered columns and numbered rows.
  - The grid itself consists of cells, and each cell can be located using unique combination of column and rows.

Database Management Software

- Used to organize, manipulate and analyze data.
- A database can contain one or more tables as well as other elements.
- In database terminology each row is a record and column is a field.
- Functions can be follows
  - Store data
  - Update data
  - Retrieve data
  - Report on data in a variety of ways
  - Print data in many forms



Presentation Graphics Software

- Used to display information in the form of a slide show.
- It has three major functions
  - An editor that allows text to be inserted and formatted.
  - A method for inserting and manipulating graphic images.
  - A slide show system to display the content.

Communication Software

- Provides method for communicating between computers.
- Most likely way to connect is via the Internet.
- Use a browser to access the Internet.
  - E.g. Chat, E-mail, FTP, Remote access
- Usage
  - Communicate from home to computer at any other place.
  - Access data stored in another computer in another location.
  - Updated information.

Software Suits

- Is a group of different but inter-related software programs that are combined and packaged together.
  - E.g. Ms Office
- Generally consists of two or more software programs delivered with in a single executable and installable file.
- Also known application suit.
- Cost of suite is less than purchasing individual applications.

Software Suits

- Advantages
  - Less cost than buying individual software.
  - All have similar GUI.
  - Work well together.

- Disadvantages
  - All features not used.
  - Takes a lot of disk space.

Characteristics of a Good Software

- Maintainability
  - The ease with which changes can be made to satisfy new requirements or to correct deficiencies.
  - Ability to evolve.

- Correctness
  - The degree with which software adheres to its specified requirements.

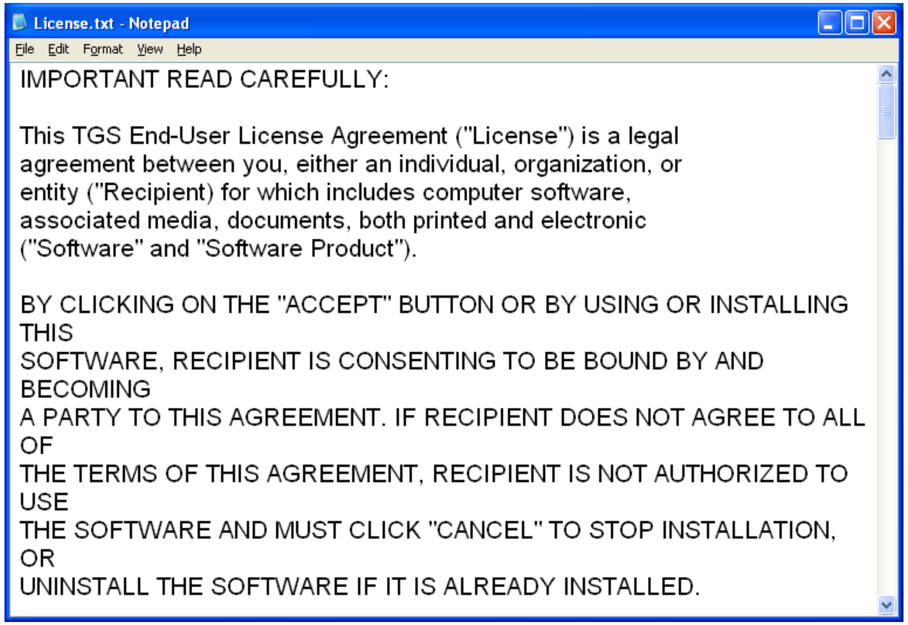
Characteristics of a Good Software

- Reusability
  - The ease with which software can be reused in developing other software.

- Reliability
  - The frequency and criticality of software failure is an unacceptable effect or behavior occurring under permissible operating conditions.

Software Licenses and Registration

- A software license gives the user the right to install and use the program on one computer.
- Organizations purchase a site license to install a program on many computers.


License.txt - Notepad

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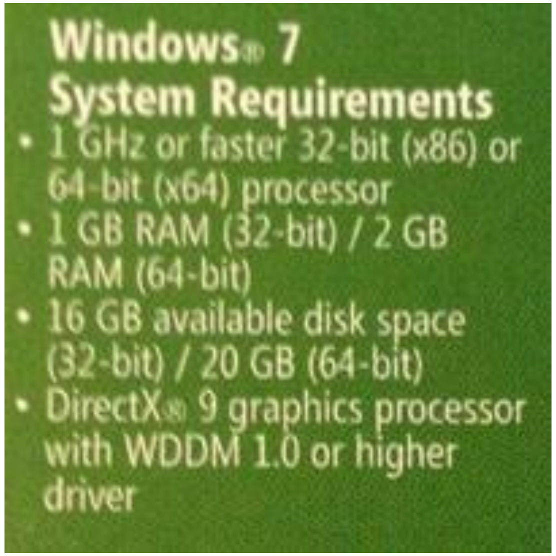
THE SOFTWARE AND MUST CLICK "CANCEL" TO STOP INSTALLATION,
OR
UNINSTALL THE SOFTWARE IF IT IS ALREADY INSTALLED.

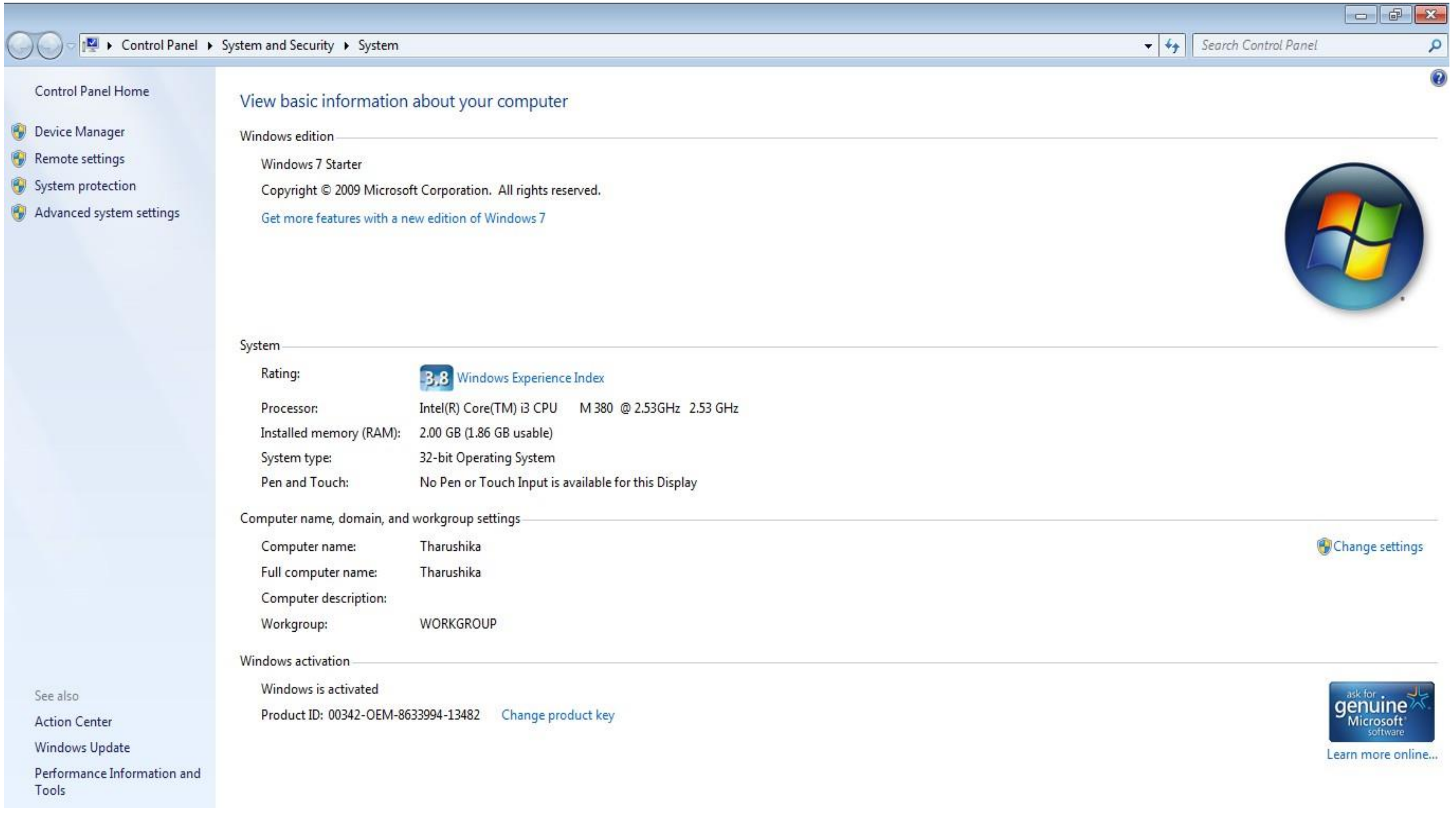
Installing and Managing Application Software

- Installing Application
  - Make sure your computer meets the system requirements of the program, game or utility you are attempting to install.
    - The system requirements are a listing of what software program or hardware devices are required to operate the program properly.



Installing and Managing Application Software

- How to check computer Specification?
  - Right click on My computer and select properties.



Installing and Managing Application Software

- How to check computer Specification?
  - Rating
    - This will only show on older versions of Windows. It is a rating of how your computer performs in multiple areas. Score range from 1.0 to 7.9, with the higher scores begin better. This is not much important.
  - Processor
    - This is the speed of your CPU. Processor model can be Intel or AMD.

Installing and Managing Application Software

- How to check computer Specification?
  - Memory (RAM)
    - This is the amount of memory you have on your computer. If you have 32 bit operating system, you can only use up to 4 GB of RAM, even if you gave more installed.
  - System type
    - This will tell you what version (32 bit or 64 bit) of Windows you are running. It also tells you if your processor is capable of running another version of Windows (32 bit (x86) or 64 bit (x64))

Installing and Managing Application Software

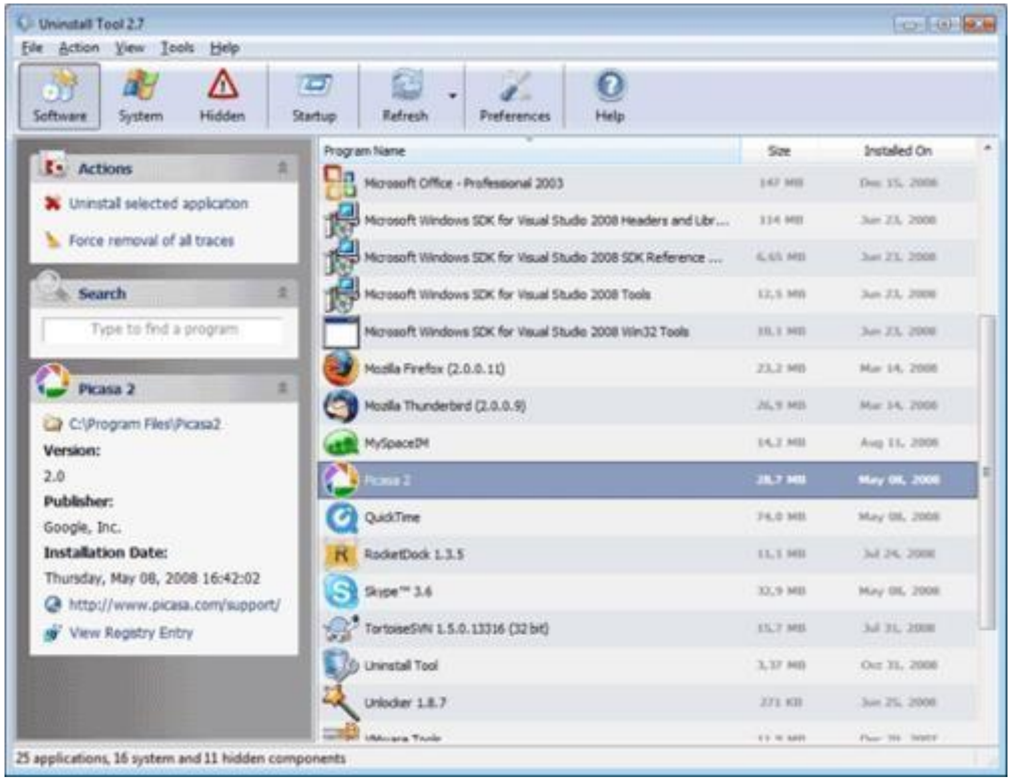
- Installing Application
  - The manual or the read me file contains exact instructions on how to install a program and are in the same directory as the installation file.
  - It is always a good idea first to close or disable any other programs that are running before installing.
  - After installing a new program, if it prompts you to reboot the computer, do it.

Installing and Managing Application Software

- Launching Application
  - Application can be launched by clicking on the Start menu, pointing to All programs, and choosing the application or selecting the desktop shortcut.
  - Launching a program transfers program code from the hard disk to the memory.
  - The program's default window appears on the screen.

Installing and Managing Application Software

- Use an uninstall utility to remove a program from the hard drive. Do not just delete a program from your files.



Quiz

- Write at least 2 examples for each of the following software types.

- Antivirus
- Audio/Music
- Database
- Game
- Email
- Graphics and Image Processing
- OS
- Browsing
- Word processing
- Spreadsheet
- Presentation

Summary


Computer Hardware
System Software
File Mngnt Tools
Operating System
Utilities
Compilers
Assembler
Debugger
Application Software
Word Processing
Spreadsheets
Graphics
Communications
Databases
Games

THANK YOU